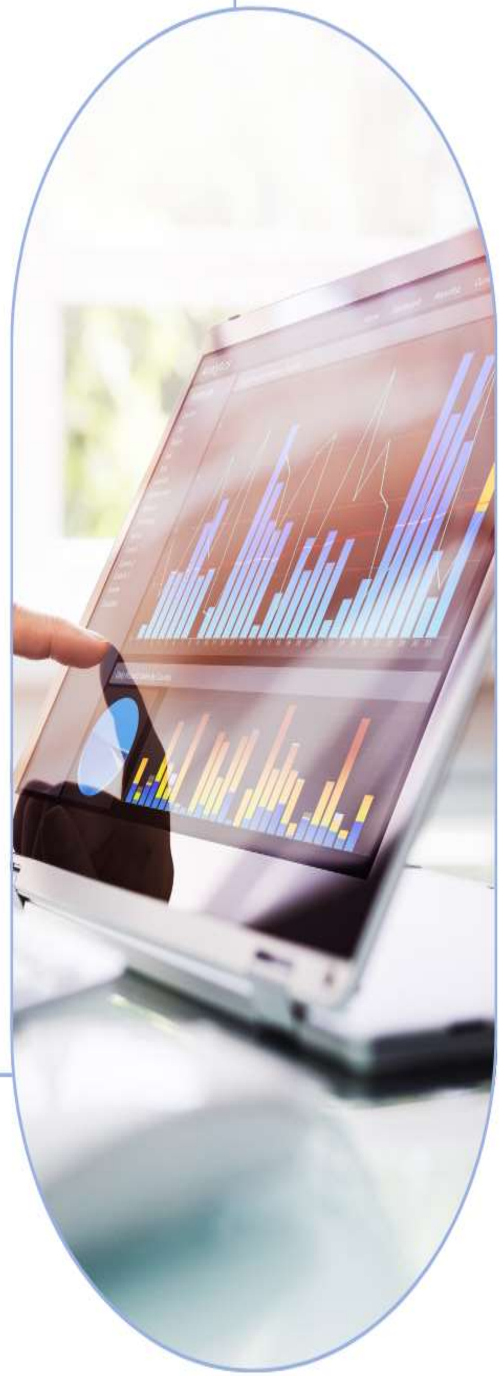
Exploring Customer Segmentation

Unveiling Insights Through Data Analysis



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Executive Summary

The primary goal of this report is to provide the Bank of Twizel with actionable insights through comprehensive customer segmentation analysis using advanced statistical analyses and machine learning techniques. By leveraging these methods, we identified five distinct customer segments based on demographic, financial, and behavioural data (Figure 1, Appendix A).

The key segments identified are Budget-Conscious Youth, Established Professionals, Growing Families, Near-Retirement Planners, and Elite Patrons. Significant differences were found between these segments in terms of age, income levels, transaction behaviours, and product ownership patterns.

These insights form the basis for developing targeted marketing and service strategies tailored to the unique needs of each segment. The primary recommendations include:

1**. Budget-Conscious Youth**: Enhance digital banking services, offer financial literacy resources and entry-level products.

2. **Established Professionals**: Introduce personalized wealth management services and investment products.

3. **Growing Families**: Provide family-centric savings plans and educational finance products.

4. **Near-Retirement Planners**: Focus on retirement planning services and secure investment options.

5. **Elite Patrons**: Offer tailored financial products for wealth transfer and ensuring financial security.

By implementing these segment-specific strategies, the Bank of Twizel can enhance customer satisfaction, drive product adoption, increase customer loyalty, and ultimately strengthen its competitive position and profitability.

A. Data-Driven Insights for Strategic Marketing

Statistical Techniques

The primary statistical technique employed in this analysis is the Two-Step Cluster Analysis. This method is particularly well-suited for the task of customer segmentation due to its ability to handle large datasets, automatically determine the optimal number of clusters, and combine both continuous and categorical variables (Tkaczynski, 2016). The quality of the resulting clusters is evaluated using statistical criteria such as the Bayesian Information Criterion (BIC) and the silhouette measure of cohesion and separation.

Additionally, Chi-Square Tests were conducted to assess the association between cluster membership and categorical variables such as product ownership and churn status. These tests help determine whether the observed differences between clusters are statistically significant.

Furthermore, Analysis of Variance (ANOVA) tests were employed to compare the means of continuous variables (e.g., income, number of transactions) across the identified clusters. This analysis helps to identify significant differences in financial behaviours among the clusters, providing valuable insights for targeted strategies.

Variable Selection

The selection of appropriate variables is crucial for effective cluster analysis and ensuring meaningful and actionable results. In this study, variables were categorized into active and passive groups.

	Active Variables	**Passive Variables**
Demographic	Income	Age
Behavioural	Transactional Activity	
	Bank Product(s) Ownership	

Table 1: Active and Passive Variables
Appendix B

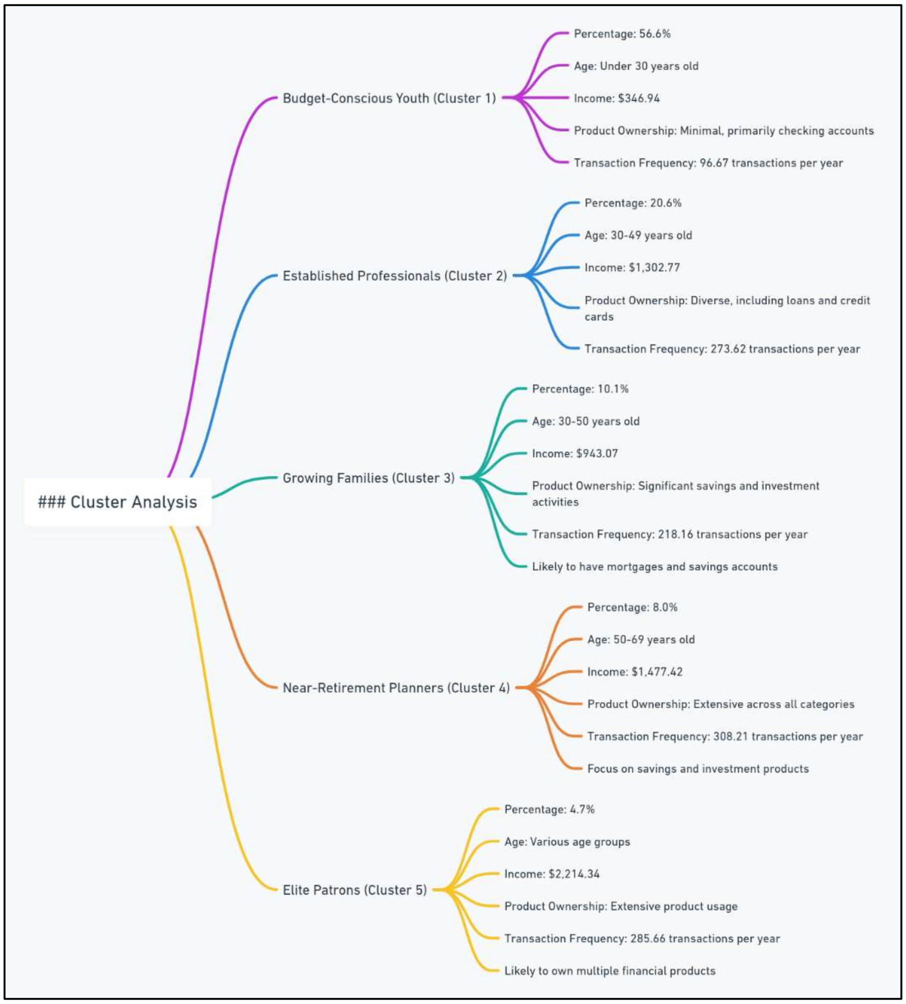
Active variables directly influence customer behaviour, financial needs, and preferences and, thus, are essential for identifying distinct customer segments. Passive variables were included to provide additional context and insights for profiling the identified customer segments (Zhang & Chang, 2020).

The selection of variables was guided by their relevance to customer segmentation and their potential impact on customer behaviour and preferences (Figure 2, Appendix A). This comprehensive set of variables ensures that the resulting clusters are meaningful, actionable, and tailored to the diverse needs of the Bank of Twizel's customer base.

B. Customer Segmentation: In-Depth Profiles and Characteristics

The Two-Step Cluster Analysis, a robust machine learning technique, successfully identified five distinct customer segments within the Bank of Twizel's customer base. The quality of the resulting clusters is supported by a high average silhouette measure of 0.7 (Figure 3, Appendix), indicating good cohesion and separation between the identified groups (Tkaczynski, 2016).

Figure A: Customer Segmentation Breakdown
Appendix B



The five segments, each with unique characteristics, are as follows:

1. **Budget-Conscious Youth (Cluster 1):** Comprising 56.6% of the customer base, this segment consists of predominantly young customers (under 30 years old) with low income ($346.94) and minimal product ownership, primarily holding checking

accounts. They exhibit low transaction frequency (96.67 transactions per year on average).

2. **Established Professionals (Cluster 2):** Representing 20.6% of customers, this segment includes middle-aged individuals (30-49 years old) with moderate income ($1,302.77) and diverse product ownership, including loans and credit cards. They engage in a moderate number of transactions (273.62 per year on average).
3. **Growing Families (Cluster 3):** Accounting for 10.1% of the customer base, this segment comprises middle-aged customers (30-50 years old) with high income ($943.07) and significant savings and investment activities. They exhibit high transaction frequencies (218.16 per year on average) and are likely to have mortgages and savings accounts.
4. **Near-Retirement Planners (Cluster 4):** Representing 8.0% of customers, this segment includes older individuals (50-69 years old) approaching retirement age with very high income ($1,477.42) and extensive product ownership across all categories. They exhibit the highest transaction volumes (308.21 per year on average) and focus on savings and investment products.
5. **Elite Patrons (Cluster 5):** The smallest segment at 4.7% of the customer base, this group comprises customers across various age groups with very high income ($2,214.34) and extensive product usage. They engage in very high transaction volumes (285.66 per year on average) and are likely to own multiple financial products.

For detailed age distribution analysis kindly refer to Figures 4 (a & b) of the Appendix A.

The Chi-Square tests conducted on categorical variables, such as product ownership and churn status, revealed significant associations with cluster membership ($p < 0.001$), (Figures 5-8, Appendix A). This underscores the distinct preferences and behaviours of each customer segment regarding financial products and services.

Furthermore, the ANOVA tests on continuous variables, including income, number of transactions, and age, demonstrated significant differences across the clusters ($p < 0.001$), please see Figures 9 (a-d), Appendix A. Games-Howell Post hoc test was used to control the Type I error rate and provides detailed insights into the specific differences between clusters, further validating the distinct profiles of each segment (Figure 10, Appendix A).

These findings highlight the effectiveness of the Two-Step Cluster Analysis in identifying meaningful customer segments based on their demographic, financial, and behavioural characteristics. The comprehensive analysis and evaluation of the results provide a solid foundation for developing targeted strategies tailored to the unique needs and preferences of each customer segment.

C. Segment-Specific Strategies: Key Recommendations

Based on the comprehensive customer segmentation analysis, the following targeted strategies are proposed as they are tailored to the unique characteristics and needs of each identified segment:

Figure B: Customer Segment Specific Strategies
Appendix B

Segment 1: Budget-Conscious Youth

Rationale: This segment comprises predominantly young customers (under 30 years old) with low income and minimal product ownership, primarily holding basic checking accounts. They exhibit low transaction frequencies and represent a significant 56.6% of the customer base, presenting an opportunity for future growth and long-term loyalty cultivation.

Recommended Strategies:

- Introduce entry-level financial products such as student accounts and low-fee credit cards to encourage more active banking engagement.
- Develop financial literacy programs and educational resources to build a strong financial foundation and foster long-term banking relationships (Resolution Life, n.d.).
- Leverage digital marketing channels and social media platforms to reach this tech-savvy segment and highlight the benefits of starting their financial journey with the Bank.

Segment 2: Established Professionals

Rationale: This segment consists of middle-aged customers (30-49 years old) with moderate income levels and diverse product ownership, including loans and credit cards. They engage in moderate transaction volumes, suggesting the potential for increased product adoption and cross-selling opportunities.

Recommended Strategies:

- Offer personalized financial advisory services and tailored wealth management solutions to cater to their evolving financial needs.
- Promote personal loans and credit cards with flexible repayment options to support their lifestyle needs.
- Leverage data-driven insights to provide personalized product recommendations through targeted communication channels.

Segment 3: Growing Families

Rationale: This segment comprises middle-aged customers (30-50 years old) with high income levels and significant savings and investment activities. They exhibit high transaction volumes, indicating a need for convenient banking experiences and family-centric financial solutions.

Recommended Strategies:

- Develop family-oriented financial products such as joint savings accounts, education savings plans, and insurance packages.
- Enhance digital banking platforms to facilitate high transaction volumes and provide seamless banking experiences.
- Offer competitive mortgage rates and family-oriented loan products to assist with major life expenses.

Segment 4: Near-Retirement Planners

Rationale: This segment includes older individuals (50-69 years old) with very high-income levels and extensive product ownership across all categories. They engage in high transaction volumes and are likely focused on retirement planning and securing their financial future.

Recommended Strategies:

- Expand retirement planning services, offering tailored advice and secure investment options to help them prepare for retirement.
- Introduce loyalty programs that reward long-term savings and investments, fostering customer retention.
- Provide specialized products and services catering to health and long-term care needs as they approach retirement.

Segment 5: Elite Patrons

Rationale: This segment comprises customers across various age groups with very high-income levels and extensive product usage. They exhibit the highest transaction volumes, indicating a need for premium financial services and personalized attention.

Recommended Strategies:

- Offer exclusive banking products and services, such as private banking, wealth management, and bespoke financial solutions.
- Implement a VIP-style service model with personalized offers, premium customer service, and direct access to senior financial advisors.
- Develop tailored marketing campaigns highlighting the benefits of advanced financial products and services.

By implementing these segment-specific strategies, the Bank of Twizel can effectively address the diverse needs of its customer base, enhance customer satisfaction and loyalty, and drive overall business growth. These recommendations are grounded in the comprehensive analysis of customer segments, their unique characteristics, financial behaviours, and product preferences, ensuring that the proposed strategies are tailored, actionable, and likely to yield positive results.

Acknowledgments:

- I acknowledge that I have cleaned the data to treat the outliers present in the data, namely, Income and Number of transactions (Figure 11-13, Appendix A).
- I acknowledge the help of Claude 3 Sonnet (Anthropic, 2024) in terms of refining, and editing the paragraphs of the assignment.

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Appendix A

Figure 1: Two Step Cluster Analysis Table (Segmentation of Bank of Twizel's Customers) Source: Bank of Twizel Dataset; IBM SPSS Software

Clusters

Input (Predictor) Importance

1.0 0.8 0.6 0.4 0.2 0.0

Cluster	**1**	**3**	**2**	**4**	**5**
Label					
Description					
Size	56.6% (31023)	20.6% (11259)	10.1% (5512)	8.0% (4399)	4.7% (2591)
Inputs	churn_t1 No (100.0%)	churn_t1 No (100.0%)	churn_t1 No (75.2%)	churn_t1 No (100.0%)	churn_t1 No (99.9%)
	Credit card 96 No (100.0%)	Credit card 96 No (100.0%)	Credit card 96 No (78.4%)	Credit card 96 Yes (100.0%)	Credit card 96 No (70.7%)
	Income 96 346.94	Income 96 943.07	Income 96 1,302.77	Income 96 1,477.42	Income 96 2,214.34
	Inv Trust 96 No (100.0%)	Inv Trust 96 No (100.0%)	Inv Trust 96 No (96.7%)	Inv Trust 96 No (100.0%)	Inv Trust 96 Yes (93.5%)
	Loan96 No (100.0%)	Loan96 No (100.0%)	Loan96 No (55.1%)	Loan96 No (100.0%)	Loan96 No (95.8%)
	Mortgage 96 No (100.0%)	Mortgage 96 No (100.0%)	Mortgage 96 No (72.2%)	Mortgage 96 No (100.0%)	Mortgage 96 No (99.0%)
	Number transactions 96 96.67	Number transactions 96 218.16	Number transactions 96 273.62	Number transactions 96 308.21	Number transactions 96 285.66
	Savings 96 No (100.0%)	Savings 96 Yes (100.0%)	Savings 96 No (64.0%)	Savings 96 Yes (51.3%)	Savings 96 Yes (66.5%)
	Checkings Acc 96 Yes (100.0%)	Checkings Acc 96 Yes (100.0%)	Checkings Acc 96 Yes (99.6%)	Checkings Acc 96 Yes (100.0%)	Checkings Acc 96 Yes (100.0%)

Figure 2: Correlation Analysis of Income, Number of Transactions and Age to each other.
Source: Bank of Twizel Dataset; IBM SPSS Software

Correlations

		Income 96	Number transactions 96	Age 96
Income 96	Pearson Correlation	1	.661**	.133**
	Sig. (2-tailed)		<.001	<.001
	N	54784	54784	54782
Number transactions 96	Pearson Correlation	.661**	1	.085**
	Sig. (2-tailed)	<.001		<.001
	N	54784	54785	54783
Age 96	Pearson Correlation	.133**	.085**	1
	Sig. (2-tailed)	<.001	<.001	
	N	54782	54783	54783

**. Correlation is significant at the 0.01 level (2-tailed).

Figure 3: Silhouette measure of cohesion and separation; Two Step Cluster Analysis
Source: Bank of Twizel Dataset; IBM SPSS Software

Model Summary

Algorithm	TwoStep
Inputs	9
Clusters	5

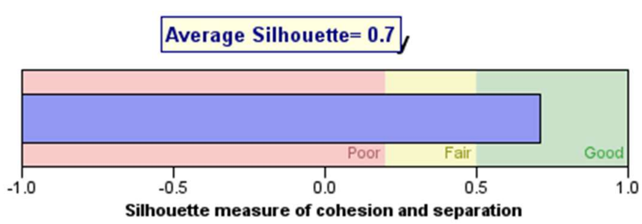

Figure 4.a: Categorical age distribution across clusters
Source: Bank of Twizel Dataset; IBM SPSS Software

Cat_Age_t0 * TwoStep Cluster Number Crosstabulation

		TwoStep Cluster Number											
		1		2		3		4		5		Total	
		N	%	N	%	N	%	N	%	N	%	N	%
Cat_Age_t0	<18	3313	10.7%	0	0.0%	182	1.6%	0	0.0%	21	0.8%	3516	6.4%
	20-29	6330	20.4%	821	14.9%	2065	18.3%	1036	23.6%	196	7.6%	10448	19.1%
	30-39	5460	17.6%	1288	23.4%	1897	16.8%	1329	30.2%	551	21.3%	10525	19.2%
	40-49	5000	16.1%	1547	28.1%	1752	15.6%	959	21.8%	589	22.7%	9847	18.0%
	50-59	3844	12.4%	943	17.1%	1444	12.8%	641	14.6%	511	19.7%	7383	13.5%
	60-69	2927	9.4%	365	6.6%	1523	13.5%	320	7.3%	396	15.3%	5531	10.1%
	70-79	2336	7.5%	237	4.3%	1561	13.9%	95	2.2%	255	9.8%	4484	8.2%
	80+	1813	5.8%	311	5.6%	835	7.4%	19	0.4%	72	2.8%	3050	5.6%
Total		31023	100.0%	5512	100.0%	11259	100.0%	4399	100.0%	2591	100.0%	54784	100.0%

Figure 4.b: Categorical age distribution across clusters
Source: Bank of Twizel Dataset; IBM SPSS Software

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	4798.932[a]	28	<.001
Likelihood Ratio	5655.682	28	<.001
Linear-by-Linear Association	372.841	1	<.001
N of Valid Cases	54784		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 144.25.

Figure 5.a: Chi-Square test for Churn and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Crosstab

		TwoStep Cluster Number											
		1		2		3		4		5		Total	
		N	%	N	%	N	%	N	%	N	%	N	%
churn_t1	No	31023	100.0%	4143	75.2%	11259	100.0%	4399	100.0%	2589	99.9%	53413	97.5%
	Yes	0	0.0%	1369	24.8%	0	0.0%	0	0.0%	2	0.1%	1371	2.5%
Total		31023	100.0%	5512	100.0%	11259	100.0%	4399	100.0%	2591	100.0%	54784	100.0%

Figure 5.b: Chi-Square test for Churn and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	12529.314[a]	4	<.001
Likelihood Ratio	6607.512	4	<.001
Linear-by-Linear Association	3.644	1	.056
N of Valid Cases	54784		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 64.84.

Figure 6.a: Chi-Square test for Product Ownership (Loan) and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Crosstab

		TwoStep Cluster Number											
		1		2		3		4		5		Total	
		N	%	N	%	N	%	N	%	N	%	N	%
Loan96	No	31023	100.0%	3036	55.1%	11259	100.0%	4399	100.0%	2483	95.8%	52200	95.3%
	Yes	0	0.0%	2476	44.9%	0	0.0%	0	0.0%	108	4.2%	2584	4.7%
Total		31023	100.0%	5512	100.0%	11259	100.0%	4399	100.0%	2591	100.0%	54784	100.0%

Figure 6.b: Chi-Square test for Product Ownership (Loan) and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	22136.051[a]	4	<.001
Likelihood Ratio	12345.465	4	<.001
Linear-by-Linear Association	60.376	1	<.001
N of Valid Cases	54784		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 122.21.

Figure 7.a: Chi-Square test for Product Ownership (Savings) and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Crosstab

		TwoStep Cluster Number											
		1		2		3		4		5		Total	
		N	%	N	%	N	%	N	%	N	%	N	%
Savings 96	No	31023	100.0%	3530	64.0%	0	0.0%	2144	48.7%	868	33.5%	37565	68.6%
	Yes	0	0.0%	1982	36.0%	11259	100.0%	2255	51.3%	1723	66.5%	17219	31.4%
Total		31023	100.0%	5512	100.0%	11259	100.0%	4399	100.0%	2591	100.0%	54784	100.0%

Figure 7.b: Chi-Square test for Product Ownership (Savings) and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	41116.571[a]	4	<.001
Likelihood Ratio	51605.968	4	<.001
Linear-by-Linear Association	26909.709	1	<.001
N of Valid Cases	54784		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 814.37.

Figure 8.a: Chi-Square test for Product Ownership (Checkings) and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

		Crosstab											
		TwoStep Cluster Number											
		1		2		3		4		5		Total	
		N	%	N	%	N	%	N	%	N	%	N	%
Checkings Acc 96	No	0	0.0%	23	0.4%	0	0.0%	0	0.0%	0	0.0%	23	0.0%
	Yes	31023	100.0%	5489	99.6%	11259	100.0%	4399	100.0%	2591	100.0%	54761	100.0%
Total		31023	100.0%	5512	100.0%	11259	100.0%	4399	100.0%	2591	100.0%	54784	100.0%

Figure 8.b: Chi-Square test for Product Ownership (Checkings) and Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	205.684[a]	4	<.001
Likelihood Ratio	105.724	4	<.001
Linear-by-Linear Association	.052	1	.820
N of Valid Cases	54784		

a. 4 cells (40.0%) have expected count less than 5. The minimum expected count is 1.09.

Figure 9.a: ANOVA Test (Oneway) for Income, Number of Transactions and Age against Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Descriptives

		N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean: Lower Bound	95% Confidence Interval for Mean: Upper Bound	Minimum	Maximum
Income 96	1	31023	346.94	545.495	3.097	340.87	353.01	0	4137
	2	5512	1302.77	1142.607	15.390	1272.60	1332.94	0	8740
	3	11259	943.07	791.244	7.457	928.45	957.69	0	5073
	4	4399	1477.42	1139.836	17.186	1443.72	1511.11	0	7420
	5	2591	2214.34	3211.492	63.092	2090.62	2338.06	0	34481
	Total	54784	744.71	1137.840	4.861	735.19	754.24	0	34481
Number transactions 96	1	31023	96.67	107.633	.611	95.47	97.87	0	625
	2	5512	273.62	194.429	2.619	268.48	278.75	0	1467
	3	11259	218.16	138.330	1.304	215.60	220.71	0	960
	4	4399	308.21	158.577	2.391	303.52	312.90	0	1205
	5	2591	285.66	186.289	3.660	278.49	292.84	0	1631
	Total	54784	165.36	157.169	.671	164.05	166.68	0	1631
Age 96	1	31023	43.01	20.538	.117	42.79	43.24	0	101
	2	5511	45.94	16.252	.219	45.51	46.37	18	101
	3	11259	49.63	19.976	.188	49.26	50.00	5	101
	4	4398	40.24	13.314	.201	39.85	40.64	18	93
	5	2591	49.73	15.556	.306	49.13	50.32	11	97
	Total	54782	44.76	19.562	.084	44.60	44.93	0	101

Figure 9.b: ANOVA Test (Oneway) for Income, Number of Transactions and Age against Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Tests of Homogeneity of Variances					
		Levene Statistic	df1	df2	Sig.
Income 96	Based on Mean	2365.717	4	54779	<.001
	Based on Median	1829.678	4	54779	<.001
	Based on Median and with adjusted df	1829.678	4	8226.461	<.001
	Based on trimmed mean	1898.025	4	54779	<.001
Number transactions 96	Based on Mean	1255.121	4	54779	<.001
	Based on Median	1136.465	4	54779	<.001
	Based on Median and with adjusted df	1136.465	4	49472.922	<.001
	Based on trimmed mean	1248.622	4	54779	<.001
Age 96	Based on Mean	580.179	4	54777	<.001
	Based on Median	539.220	4	54777	<.001
	Based on Median and with adjusted df	539.220	4	52682.103	<.001
	Based on trimmed mean	575.333	4	54777	<.001

Figure 9.c: ANOVA Test (Oneway) for Income, Number of Transactions and Age against Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

ANOVA						
		Sum of Squares	df	Mean Square	F	Sig.
Income 96	Between Groups	15025873512	4	3756468377.9	3681.097	<.001
	Within Groups	55900616702	54779	1020475.304		
	Total	70926490213	54783			
Number transactions 96	Between Groups	369633193.81	4	92408298.452	5146.354	<.001
	Within Groups	983615564.14	54779	17956.070		
	Total	1353248757.9	54783			
Age 96	Between Groups	522722.787	4	130680.697	350.216	<.001
	Within Groups	20439682.691	54777	373.144		
	Total	20962405.478	54781			

Figure 9.d: ANOVA Test (Oneway) for Income, Number of Transactions and Age against Cluster Members
Source: Bank of Twizel Dataset; IBM SPSS Software

Robust Tests of Equality of Means					
		Statistic[a]	df1	df2	Sig.
Income 96	Welch	3164.598	4	9553.231	<.001
Number transactions 96	Welch	4431.774	4	9890.256	<.001
Age 96	Welch	421.982	4	11376.679	<.001

a. Asymptotically F distributed.

Figure 10: Post Hoc Test (Games-Howell)
Source: Bank of Twizel Dataset; IBM SPSS Software

Multiple Comparisons

Games-Howell

Dependent Variable	(I) TwoStep Cluster Number	(J) TwoStep Cluster Number	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	Upper Bound
Income 96	1	2	-955.829*	15.699	<.001	-998.66	-912.99
		3	-596.134*	8.075	<.001	-618.16	-574.11
		4	-1130.480*	17.462	<.001	-1178.13	-1082.83
		5	-1867.403*	63.168	<.001	-2039.83	-1694.97
	2	1	955.829*	15.699	<.001	912.99	998.66
		3	359.695*	17.102	<.001	313.04	406.35
		4	-174.650*	23.069	<.001	-237.59	-111.71
		5	-911.574*	64.942	<.001	-1088.83	-734.32
	3	1	596.134*	8.075	<.001	574.11	618.16
		2	-359.695*	17.102	<.001	-406.35	-313.04
		4	-534.346*	18.734	<.001	-585.46	-483.23
		5	-1271.269*	63.531	<.001	-1444.69	-1097.85
	4	1	1130.480*	17.462	<.001	1082.83	1178.13
		2	174.650*	23.069	<.001	111.71	237.59
		3	534.346*	18.734	<.001	483.23	585.46
		5	-736.923*	65.391	<.001	-915.40	-558.44
	5	1	1867.403*	63.168	<.001	1694.97	2039.83
		2	911.574*	64.942	<.001	734.32	1088.83
		3	1271.269*	63.531	<.001	1097.85	1444.69
		4	736.923*	65.391	<.001	558.44	915.40
Number transactions 96	1	2	-176.949*	2.689	<.001	-184.29	-169.61
		3	-121.490*	1.440	<.001	-125.42	-117.56
		4	-211.542*	2.468	<.001	-218.28	-204.81
		5	-188.994*	3.710	<.001	-199.12	-178.87
	2	1	176.949*	2.689	<.001	169.61	184.29
		3	55.459*	2.925	<.001	47.48	63.44
		4	-34.593*	3.546	<.001	-44.27	-24.92
		5	-12.045	4.500	.058	-24.33	.23
	3	1	121.490*	1.440	<.001	117.56	125.42
		2	-55.459*	2.925	<.001	-63.44	-47.48
		4	-90.052*	2.723	<.001	-97.48	-82.62
		5	-67.504*	3.885	<.001	-78.11	-56.90
	4	1	211.542*	2.468	<.001	204.81	218.28
		2	34.593*	3.546	<.001	24.92	44.27
		3	90.052*	2.723	<.001	82.62	97.48
		5	22.548*	4.372	<.001	10.62	34.48
	5	1	188.994*	3.710	<.001	178.87	199.12
		2	12.045	4.500	.058	-.23	24.33
		3	67.504*	3.885	<.001	56.90	78.11
		4	-22.548*	4.372	<.001	-34.48	-10.62
Age 96	1	2	-2.926*	.248	<.001	-3.60	-2.25
		3	-6.614*	.221	<.001	-7.22	-6.01
		4	2.772*	.232	<.001	2.14	3.41
		5	-6.711*	.327	<.001	-7.60	-5.82
	2	1	2.926*	.248	<.001	2.25	3.60
		3	-3.688*	.289	<.001	-4.48	-2.90
		4	5.698*	.297	<.001	4.89	6.51
		5	-3.785*	.376	<.001	-4.81	-2.76
	3	1	6.614*	.221	<.001	6.01	7.22
		2	3.688*	.289	<.001	2.90	4.48
		4	9.386*	.275	<.001	8.64	10.14
		5	-.097	.359	.999	-1.08	.88
	4	1	-2.772*	.232	<.001	-3.41	-2.14
		2	-5.698*	.297	<.001	-6.51	-4.89
		3	-9.386*	.275	<.001	-10.14	-8.64
		5	-9.483*	.366	<.001	-10.48	-8.48
	5	1	6.711*	.327	<.001	5.82	7.60
		2	3.785*	.376	<.001	2.76	4.81
		3	.097	.359	.999	-.88	1.08
		4	9.483*	.366	<.001	8.48	10.48

*. The mean difference is significant at the 0.05 level.

Figure 11: Outlier Visualisation for Age (Boxplot)
Source: Bank of Twizel Dataset; IBM SPSS Software

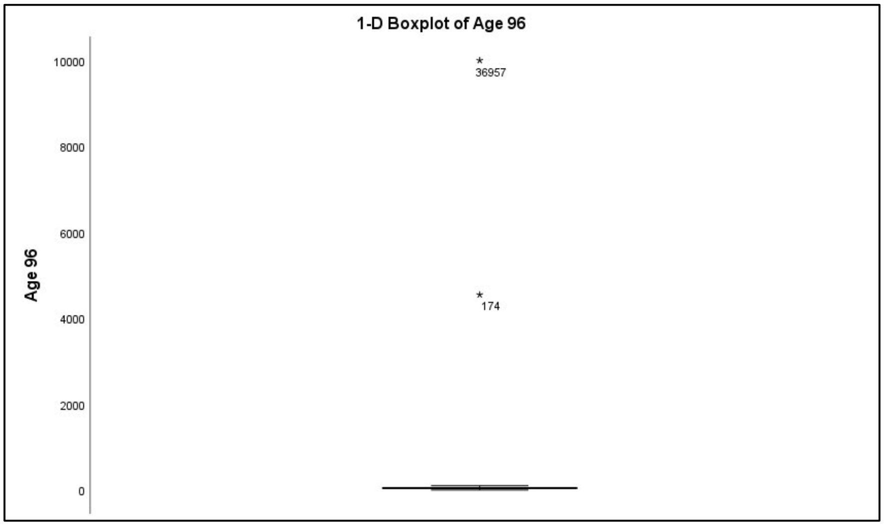

Figure 12: Outlier Visualisation for income (Boxplot)
Source: Bank of Twizel Dataset; IBM SPSS Software

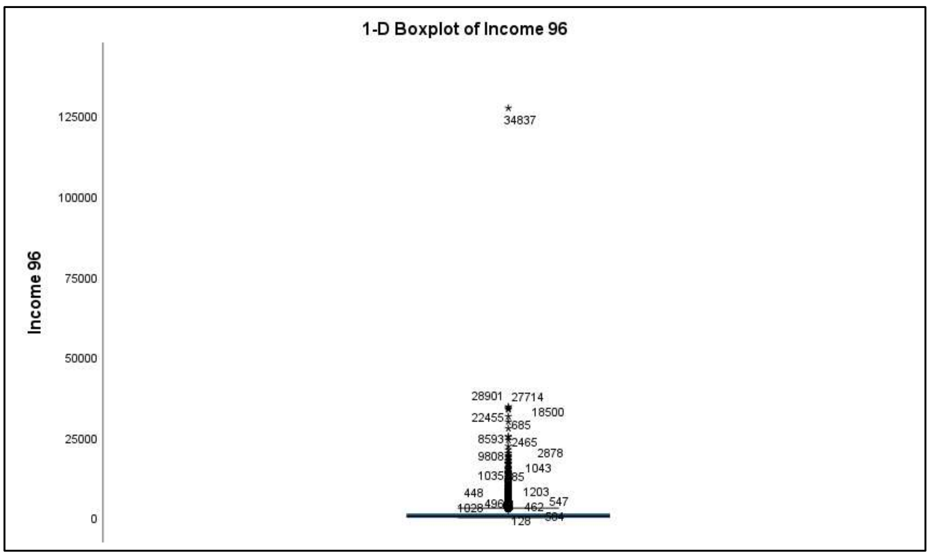

Figure 13: Statistical Summary (Cleaned data Output)
Source: Bank of Twizel Dataset; IBM SPSS Software

Statistics

		Income 96	Age 96
N	Valid	54784	54783
	Missing	1	2
Mean		744.71	44.76
Median		454.00	42.00
Mode		0	50
Std. Deviation		1137.840	19.562
Range		34481	101

Appendix B

Table 1: Active and Passive Variables

	Active Variables	Passive Variables
Demographic	Income	Age
Behavioural	Transactional Activity	
	Bank Product(s) Ownership	

Figure A: Customer Segmentation Breakdown

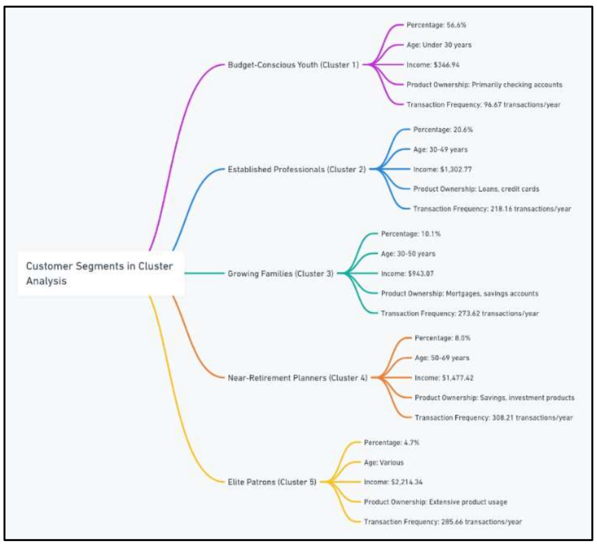

Figure B: Customer Segment Specific Strategies